# A kinematic model that predicts joint movements to improve the gameplay of virtual reality

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People Interfaces +Data

## Background

Virtual reality (VR) has become increasingly prevalent in our daily lives, with applications spanning from military training to medical surgery, each benefiting from VR's highly immersive nature. Currently, VR systems primarily rely on the position and orientation of the headset and controllers to track players' movements. While these readings lead to a smooth gameplay overall, complete knowledge of upper limb joint movements will allow the development of new interactions with a virtual environment. By tracking and modeling movements of the upper body joint in VR, not only can we improve recreational VR use, but allow VR-mediated physical therapy to have more accurate data, thereby helping healthcare providers and patients alike.

## Limitations of Virtual Reality

Although VR technology is consistently advancing, the position and orientation of the controllers and head-mounted display are insufficient for rehabilitation purposes.

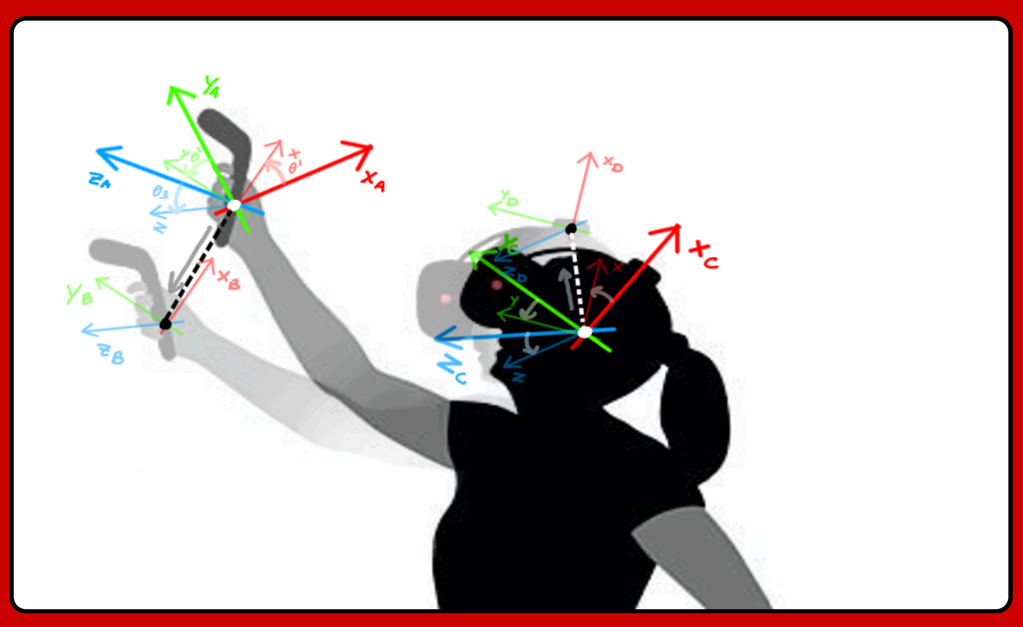


Figure 1: Illustration of a person playing a game in VR, where the position and orientation of the headset and hand controllers are represented with a reference frame.

## Application of Machine Learning

Machine learning offers a means to predict angles of the upper limb joints from data collected by a VR system. However, this approach could benefit from a kinematic model that would inform on physical relationships between the VR system and joint angles.

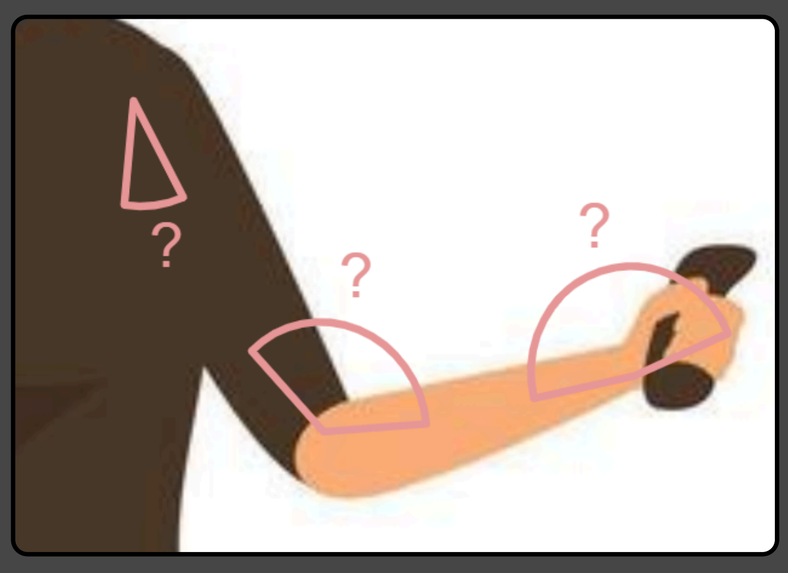



Figure 2: Illustration of angles that are currently not measured by a VR system and may be predicted with machine learning

## Kinematic model of a human arm

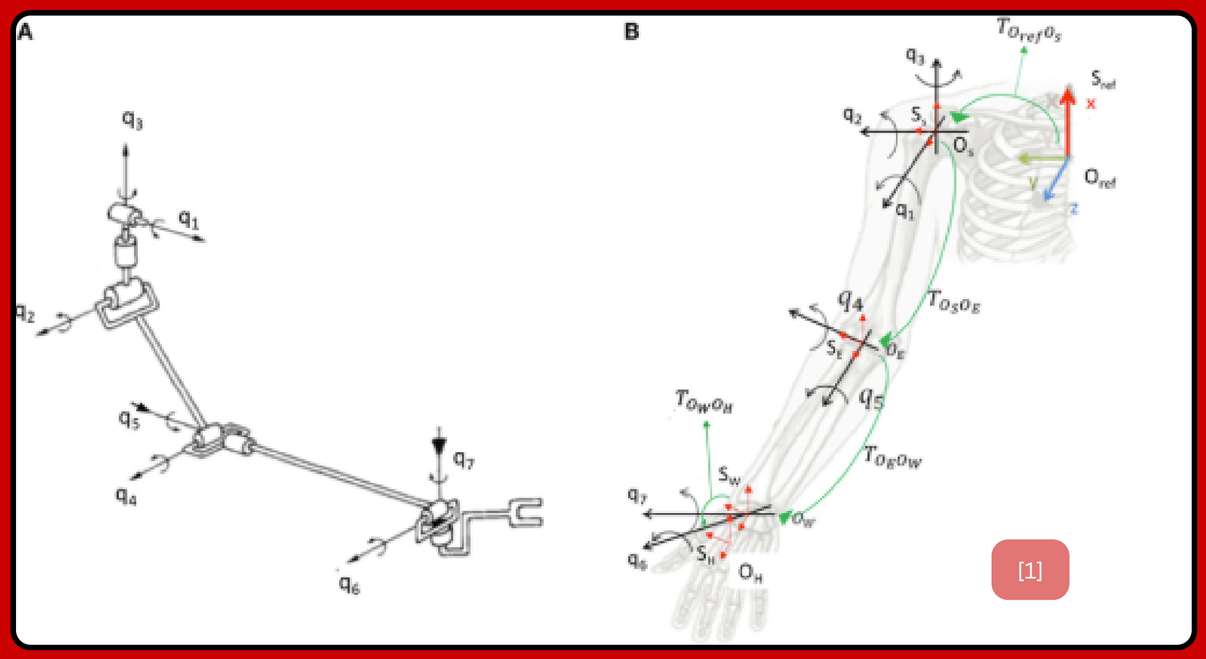


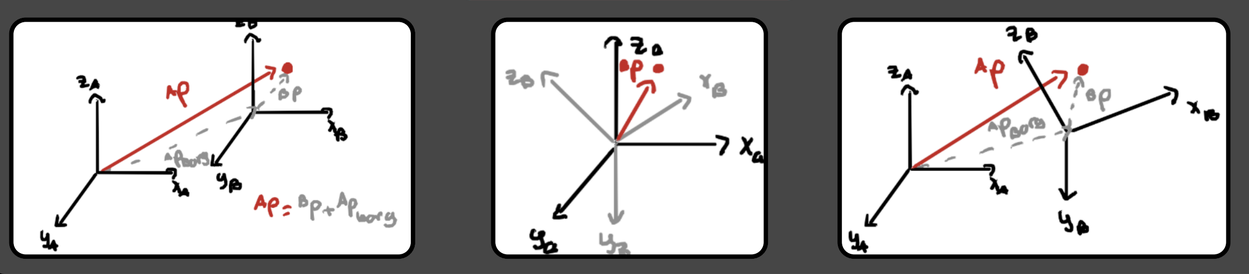

Figure 3: Model of a robotic arm (A) and a human arm (B).

## Representing kinematics using reference frames

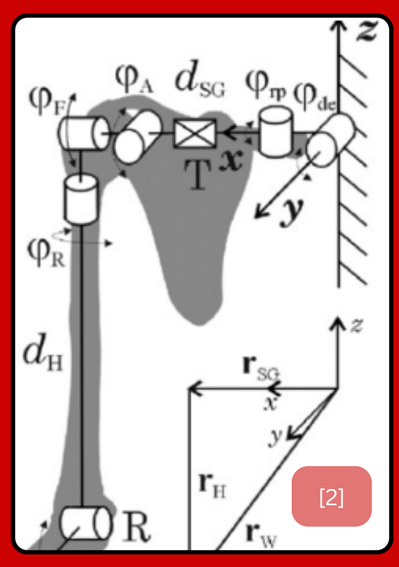
**Translation: Origin changes, orientation stays the same**

**Rotation: Origin stays the same, orientation changes**

**Both : Origin changes, orientation changes**



## Kinematic Model of the Shoulder




$\varphi_A = [\varphi_{min}, \varphi_{max}]$

$\varphi_A$: $adduction/abduction$

$\varphi_F = [\varphi_{Fmin} + \frac{\varphi_A}{3}, \varphi_{Fmax} - \frac{\varphi_A}{6}]$

$\varphi_F$: $retroflexion/flexion$

$\varphi_R = [\varphi_{Rmin} + \frac{7\varphi_A}{9} - \frac{\varphi_F}{9} + \frac{2\varphi_A\varphi_F}{810}, \varphi_{Rmax} + \frac{4\varphi_A}{9} - \frac{5\varphi_F}{9} + \frac{5\varphi_A\varphi_F}{810}]$

$\varphi_R$: $internal/external\ rotation$

$\varphi = arccos(cos\varphi_A cos\varphi_F)$

$\varphi$: $elevation\ angle\ of\ the\ humerus$

[2]

## Next Steps

In the kinematic model, we use forward and backward kinematics to produce relationships between measurements that are recorded by a VR system and joint angles. These relationships will inform features in a machine learning algorithm that predicts joints of the entire upper limb. The algorithm will be trained on data from healthy users who are playing a game in VR, while their motion is tracked by a motion capture system for validation. Being able to predict full body kinematics in VR in will advance many fields such as rehabilitation, where specialists to monitor their patient's exercises in VR-mediated physical therapy.

## References:

[1] Averta, G., et al. "Unvealing the principal modes of human upper limb movements through functional analysis." *Frontiers in Robotics and AI* 4 (2017): 37.

[2] Klopčar, N., et al. "A kinematic model of the shoulder complex to evaluate the arm-reachable workspace." *Journal of Biomechanics* 40.1 (2007): 86-91.

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